

Supplementary Information

Helium gas or liquid nitrogen for the first thermal shield of a 4 K cryostat: conjugate heat-transfer budget, coolant operating maps and work ledger

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All cases, scripts and tabulated results: github.com/Imerson/Pre-cooling-a-superconducting-qubit-dilution-refrigerator (release v2).

S1 Enclosure model

Table S1. Regions, patches and boundary conditions of the enclosure CHT cases.

Region	Patch	T condition	Radiation condition
domain0 (OVC-shield gap)	warmPlate, OVC	fixed T_{hot}	grey diffuse, $\varepsilon = 0.04$
	shield_slave	fixed T_{cold}	grey diffuse, $\varepsilon = 0.02$
domain1 (inner gap)	warmPlate	fixed T_{hot}	grey diffuse, $\varepsilon = 0.04$
	coldPlate, shield_bottom	fixed T_{cold}	grey diffuse, $\varepsilon = 0.04$
	shield	fixed T_{cold}	grey diffuse, $\varepsilon = 0.02$
	domain1_to_coax_Lk	conjugate (mapped)	grey diffuse, $\varepsilon = 0.02$
coax_L1..L4 (solids)	warm cap / cold cap	fixed T_{hot} / T_{cold}	–
	lateral	conjugate to domain1	–

Table S2. Gas-region settings that make the gaps behave as vacuum. Radiation in OpenFOAM’s **viewFactor** model is solved in fluid regions, so each gap is a fluid region made inert.

Setting	Value
pressure $p = p_{\text{rgh}}$	10^{-3} Pa (pRefValue 10^{-3})
gravity	(0, 0, 0)
transport	constant, $\mu = 10^{-11}$ Pa s, $\text{Pr} = 0.67$
density limits	$\rho_{\text{max}} = 10^{-4}$, $\rho_{\text{min}} = 10^{-12}$ kg m $^{-3}$
radiation solver frequency	every 10 iterations
agglomeration	nFacesInCoarsestLevel 250, featureAngle 20, mergeLevels 1

Table S3. Harness representation. Cable UT-085-SS-SS: 304 stainless outer conductor $A = 1.585$ mm 2 , PTFE dielectric $A = 2.001$ mm 2 , 304 stainless centre conductor $A = 0.205$ mm 2 . Material $k(T)$ from the NIST cryogenic fits $\log_{10} k = \sum_n a_n (\log_{10} T)^n$. Each modelled cylinder carries n_k lines with $\kappa_{\text{eff}}(T) = (n_k/A_{\text{model}}) \sum_i A_i k_i(T)$; A_{model} is the calibrated effective conduction area of the 20 mm cylinder.

Bundle	Family	n_k	share of harness conduction
coax_L1	drive	25	38.5 %
coax_L2	flux	25	38.5 %
coax_L3	readout	10	15.4 %
coax_L4	pump	5	7.7 %

Table S4. Deployed $\kappa_{\text{eff}}(T) = \sum_{n=0}^5 c_n T^n$ [W m⁻¹K⁻¹, T in K] per bundle and stage span (`kappaCoeffs<8>`, last two coefficients zero). Fit error < 0.5 % over each span.

Span	Bundle	c_0	c_1	c_2	c_3	c_4	c_5
300→50 K	L1	-2.1522×10^{-1}	3.1627×10^{-2}	-2.3184×10^{-4}	9.9921×10^{-7}	-2.1908×10^{-9}	1.9383×10^{-12}
	L2	-2.1519×10^{-1}	3.1622×10^{-2}	-2.3181×10^{-4}	9.9907×10^{-7}	-2.1905×10^{-9}	1.9380×10^{-12}
	L3	-8.6092×10^{-2}	1.2651×10^{-2}	-9.2741×10^{-5}	3.9970×10^{-7}	-8.7636×10^{-10}	7.7535×10^{-13}
	L4	-4.3044×10^{-2}	6.3253×10^{-3}	-4.6369×10^{-5}	1.9984×10^{-7}	-4.3816×10^{-10}	3.8766×10^{-13}
300→77 K	L1	-1.3459×10^{-1}	2.9103×10^{-2}	-2.0205×10^{-4}	8.3171×10^{-7}	-1.7398×10^{-9}	1.4706×10^{-12}
	L2	-1.3457×10^{-1}	2.9099×10^{-2}	-2.0202×10^{-4}	8.3159×10^{-7}	-1.7396×10^{-9}	1.4704×10^{-12}
	L3	-5.3837×10^{-2}	1.1642×10^{-2}	-8.0824×10^{-5}	3.3269×10^{-7}	-6.9596×10^{-10}	5.8824×10^{-13}
	L4	-2.6917×10^{-2}	5.8206×10^{-3}	-4.0410×10^{-5}	1.6634×10^{-7}	-3.4796×10^{-10}	2.9411×10^{-13}
50→4 K	L1	-4.8624×10^{-4}	9.1378×10^{-4}	3.6621×10^{-5}	-9.0343×10^{-7}	8.4798×10^{-9}	-2.8699×10^{-11}
	L2	-4.8624×10^{-4}	9.1378×10^{-4}	3.6621×10^{-5}	-9.0343×10^{-7}	8.4798×10^{-9}	-2.8699×10^{-11}
	L3	-1.9450×10^{-4}	3.6551×10^{-4}	1.4648×10^{-5}	-3.6137×10^{-7}	3.3919×10^{-9}	-1.1480×10^{-11}
	L4	-9.7247×10^{-5}	1.8275×10^{-4}	7.3242×10^{-6}	-1.8068×10^{-7}	1.6959×10^{-9}	-5.7398×10^{-12}
77→4 K	L1	-7.4665×10^{-4}	1.2814×10^{-3}	4.9472×10^{-5}	-1.2245×10^{-6}	1.1714×10^{-8}	-4.2617×10^{-11}
	L2	-7.4665×10^{-4}	1.2814×10^{-3}	4.9472×10^{-5}	-1.2245×10^{-6}	1.1714×10^{-8}	-4.2617×10^{-11}
	L3	-2.9865×10^{-4}	5.1254×10^{-4}	1.9788×10^{-5}	-4.8978×10^{-7}	4.6855×10^{-9}	-1.7046×10^{-11}
	L4	-1.4932×10^{-4}	2.5626×10^{-4}	9.8938×10^{-6}	-2.4488×10^{-7}	2.3427×10^{-9}	-8.5230×10^{-12}

Free-molecular gas conduction bound. At $p < 10^{-4}$ mbar the mean free path exceeds the gap and gas heat transport is free-molecular, $\dot{Q}_{\text{gas}} = \alpha \Lambda p A \Delta T$ [1]; it is linear in pressure and below 1 % of the harness conduction at the operating pressure, so it is carried as a budget line.

Table S5. View-factor agglomeration refinement and row normalisation. Closure = mean row sum $\sum_j F_{ij}$ of the generated matrix (unity for a closed enclosure); loads with the matrix as generated and, at the production level, with rows normalised.

Case	Coarsest-level faces	Closure (mean; range)	Radiation, as generated	Radiation, normalised
300→50 K 60		–	9.46 W	–
300→50 K 250 (production)		0.99–1.01; 0.50–1.13	8.41 W	7.36 W
300→50 K 400		0.98–0.99	6.49 W	–
300→50 K 800		–	3.26 W	–
300→50 K closed form		–	7.07 W	–
300→77 K 250 (production)		0.99	8.38 W	7.34 W (closed form 7.04)
50→4 K 250 (production)		1.00	5.29 mW	3.16 mW (closed form 3.15)
77→4 K 250 (production)		1.00	29.7 mW	17.8 mW (closed form 17.7)

Row normalisation. Each fluid region of the enclosure model is a closed surface, so every row of the view-factor matrix must sum to unity. The generated matrix (`viewFactorsGen`, Gauss quadrature tolerance 0.01) is stored sparsely, one entry per visible face pair; the normalisation divides each row by its sum and leaves rows with a sum below 0.2 (none occurred) untouched. The solve was then restarted from the converged production field for 600 further iterations. Script `normalise_F.py` in the repository.

Table S6. Emissivity sensitivity (production view factors) and MLI-equivalent variant.

Case	ε warm / cold	Radiation [W]	Total [W]
300→50 K, low	0.02 / 0.01	5.45	6.48
300→50 K, baseline	0.04 / 0.02	8.41	9.44
300→50 K, high	0.06 / 0.03	11.48	12.51
300→77 K, low	0.02 / 0.01	5.44	6.40
300→77 K, baseline	0.04 / 0.02	8.38	9.35
300→77 K, high	0.06 / 0.03	11.44	12.40
300→50 K, MLI-eq.	0.04 / 0.003	1.81	2.84
300→77 K, MLI-eq.	0.04 / 0.003	1.81	2.77

S2 Cold-plate sweep: full results

Table S7. All cold-plate runs at 9.4 W (production mesh). $\dot{m}$ in g s^{-1} , volumetric flow in $\text{cm}^3 \text{s}^{-1}$, Δp in Pa, $\dot{W}_p$ in μW , temperatures in K, h in $\text{W m}^{-2}\text{K}^{-1}$; $T_{\text{in}} = 45 \text{ K}$ (He), 77 K (LN_2); saturation margin at 3 bar; y^+ = maximum on the heated wall; closure = energy-balance error. Gravity-on (g on) nitrogen runs did not reach a steady state (see closure).

Case	Re	$\dot{m}$	$\dot{V}$	Δp	$\dot{W}_p$	T_{out}	T_w	$T_{w,\text{max}}$	$T_w - T_b$	h	Nu	f	NTU	ε	Gr/ Re^2	margin	y^+	closure %
He 1 bar $Re=500$	500	0.159	167	0.294	48.6	56.36	87.4	97.6	36.72	17.1	3.66	0.186	0.310	0.266	0.63	–	–	-0.06
He 1 bar $Re=500$ (g on)	500	0.159	167	0.331	54.6	56.21	67.5	48.4	16.87	37.1	7.96	0.21	0.674	0.490	0.29	–	–	-1.41
He 1 bar $Re=1000$	1000	0.318	316	0.671	211	50.68	76.8	83.0	28.94	21.7	4.64	0.112	0.196	0.178	0.15	–	–	-0.05
He 1 bar $Re=1500$	1500	0.477	465	1.12	521	48.79	71.7	75.8	24.77	25.3	5.42	0.0847	0.153	0.142	0.06	–	–	-0.05
He 1 bar $Re=2300$	2300	0.731	702	1.98	1.39e+03	47.47	67.1	69.4	20.82	30.1	6.45	0.0642	0.119	0.112	0.022	–	–	-0.06
He 1 bar $Re=2300$ (g on)	2300	0.731	703	2.02	1.42e+03	47.48	62.7	59.3	16.41	38.2	8.18	0.0656	0.151	0.140	0.018	–	–	0.55
He 1 bar $Re=5000$	5000	1.59	1.5e+03	6.62	9.95e+03	46.14	55.3	51.9	9.73	64.4	13.8	0.0461	0.117	0.110	0.0023	–	2.9	-0.08
He 1 bar $Re=10000$	10000	3.18	2.99e+03	21.8	6.51e+04	45.57	50.3	47.9	5.00	125	26.8	0.0382	0.114	0.107	0.0003	–	5.6	-0.18
He 5 bar $Re=2300$	2300	0.731	140	0.395	55.5	47.47	67.1	69.4	20.82	30.1	6.45	0.0641	0.119	0.112	0.56	–	–	-0.04
He 18 bar $Re=2300$	2300	0.731	39	0.11	4.29	47.47	67.1	69.4	20.82	30.1	6.45	0.0643	0.119	0.112	7.3	–	–	-0.04
LN_2 3 bar $Re=500$	500	4.09	5.06	0.202	1.02	78.13	85.0	85.7	7.46	84	5.77	0.163	0.151	0.140	40	2.2	–	0.00
LN_2 3 bar $Re=500$ (g on)	500	4.09	5.06	1.68	8.5	77.32	78.9	78.6	1.74	361	24.8	1.35	0.650	0.478	9.4	9.3	–	-71.96
LN_2 3 bar $Re=1000$	1000	8.17	10.1	0.501	5.07	77.56	83.1	83.1	5.77	109	7.45	0.101	0.098	0.093	7.8	4.8	–	0.00
LN_2 3 bar $Re=1500$	1500	12.3	15.2	0.869	13.2	77.38	82.1	81.9	4.92	127	8.75	0.0778	0.076	0.074	2.9	6.0	–	0.00
LN_2 3 bar $Re=2300$	2300	18.8	23.3	1.57	36.6	77.25	81.2	80.8	4.12	152	10.4	0.0599	0.060	0.058	1	7.1	–	-0.00
LN_2 3 bar $Re=2300$ (g on)	2300	18.8	23.3	3	69.8	77.22	78.5	77.6	1.35	463	31.8	0.114	0.181	0.166	0.34	10.4	–	-10.77
LN_2 3 bar $Re=5000$	5000	40.9	50.6	5.57	282	77.11	78.6	77.6	1.55	405	27.8	0.0449	0.073	0.070	0.083	10.3	2.9	0.04
LN_2 3 bar $Re=10000$	10000	81.7	101	18.6	1.88e+03	77.06	77.8	77.3	0.81	777	53.3	0.0374	0.070	0.068	0.011	10.6	5.6	0.03

Table S8. Grid convergence index study for both fluids at $Re = 2300$ (Celik et al. 2008); meshes 86 400 / 291 600 / 984 150 cells, refinement ratio 1.5, production mesh = medium.

Fluid	Quantity	Coarse	Medium	Fine	p	Richardson	$\text{GCI}_{\text{medium}}$	GCI_{fine}
He, 1 bar	Nu	6.439	6.449	6.457	0.64	6.485	0.70 %	0.54 %
He, 1 bar	f	0.0632	0.0642	0.0647	1.40	0.0655	2.5 %	1.4 %
He, 1 bar	h [$\text{W m}^{-2}\text{K}^{-1}$]	30.05	30.10	30.14	0.64	30.27	0.70 %	0.54 %
LN_2 , 3 bar	Nu	10.513	10.436	10.382	0.87	10.26	2.2 %	1.5 %
LN_2 , 3 bar	f	0.0592	0.0599	0.0603	1.71	0.0607	1.6 %	0.8 %
LN_2 , 3 bar	h [$\text{W m}^{-2}\text{K}^{-1}$]	153.2	152.1	151.3	0.87	149.5	2.2 %	1.5 %

Property correction and energy accounting. The helium transport properties of the earlier version of this comparison ($\mu = 2.73 \mu\text{Pa s}$, $k = 0.0185 \text{ W m}^{-1}\text{K}^{-1}$) correspond to helium near 15 K; the present runs use the 50 K values ($6.36 \mu\text{Pa s}$, $0.0467 \text{ W m}^{-1}\text{K}^{-1}$, CoolProp/NIST). The mixing-cup outlet temperature is the mass-flux-weighted average $\sum \phi_f T_f / \sum \phi_f$ over the outlet faces (surfaceFieldValue, weightedAverage with phi); the weightedAreaAverage operation, which multiplies the mass flux by the face area a second time, biases the outlet temperature low on a graded mesh and was the cause of the 2–10 % energy imbalances reported earlier. The delivered wall heat flux was verified with the solver-native wallHeatFlux function object (9.395 W).

S3 Hardware efficiencies used in the real ledger

Table S9. Stage efficiencies as fractions of Carnot.

Stage	η range	Source
4 K cryocooler stage	0.008–0.015	PT415: 1.5 W at 4.2 K with 40 W at 45 K for 10.7 kW [2]; input allocated by Carnot work gives 0.031 for both stages combined; Radebaugh: “at 4 K efficiencies of about 1 % of Carnot or less are typical” [3]
He first stage (45–80 K)	0.10–0.15	Radebaugh: 10–15 % of Carnot for the best commercial 80 K coolers; large plants up to 30 % [3, 4]
LN_2 supply	0.30–0.40	ideal liquefaction work 0.213 kWh kg $^{-1}$ [5]; plant consumption 0.5–0.7 kWh kg $^{-1}$ [6]; latent heat 199 kJ kg $^{-1}$

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